DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE – RAIGAD -402 103
Supplementary Winter Examination – 2023

Branch: Electronics and Telecommunication Engineering/ Electronics and Communication Engineering
Course: B. Tech **Sem.:- VI**
Subject with Subject Code: - Digital communication BTETC602 **Marks: 60**
Date:- 18/01/2024 (2 to 5 PM) **Time:3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	Explain Quantization Noise in PCM.	L1	6
B)	What is Delta modulation? Explain its advantages and disadvantages. Explain various noises occurring in Delta Modulation.	L2	6
C)	Explain basic Digital communication system with neat block diagram.	L1	6
Q.2	**Solve Any Two of the following.**		**12**
A)	Differentiate between Pulse Code Modulation (PCM) and Differential Pulse Code Modulation (DPCM)	L3	6
B)	Write a note on Wide-sense stationary process.	L2	6
C)	Write a note on Additive scramblers. Explain its applications.	L1	6
Q. 3	**Solve Any Two of the following.**		**12**
A)	What are the cause and effects of Inter Symbol Interference? Explain Eye diagram.	L3	6
B)	Explain the working of optimum receiver system with neat block diagram. Also obtain equation for probability of error for the same.	L2	6
C)	Explain prominent features of M-ary PSK.	L2	6

Q.4	**Solve Any Two of the following.**			**12**
A)	What is the frequency range, spectral characteristics and application of narrowband noise?	L1		6
B)	What is Matched Filter? Explain its working with the help of neat block diagram.	L1		6
C)	Explain Transmission Rates used in T-Carrier system.	L2		6
Q. 5	**Solve Any Two of the following.**			**12**
A)	Explain the concept of Jammers. What are its applications?	L1		6
B)	Explain the working of Explain the working of Correlation receiver.	L2		6
C)	Explain DS-SS technique. What are its advantages and applications?	L2		6

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